Name: ____________________________

Date: ____________________________

Mr. Rawson • Y9 MYP Physics

1.1 – Oscillations and Frequency

Unit 1: Oscillations & Waves

Learning intentions

- a) I can define an **oscillation** as a repeating motion about a rest position.
- b) I can identify everyday examples of oscillating systems.
- c) I can calculate **frequency** from the number of oscillations and time taken.
- d) I can convert between hertz (Hz), kilohertz (kHz), megahertz (MHz) and gigahertz (GHz).

Theory

An **oscillation** is a repeating back-and-forth motion about a central **rest position**. The rest position is the point where an object would stay if it were not moving. One complete oscillation means the object moves away from the rest position, reaches its maximum distance, and returns to the start. Everyday examples include a swinging pendulum, a mass bouncing on a spring, a vibrating guitar string, a tuning fork, or even your heartbeat.

The speed of these vibrations is described by **frequency**. Frequency is defined as the number of complete oscillations that occur in one second. The standard unit for frequency is the **hertz** (symbol: Hz). If an object completes 50 oscillations in one second, its frequency is 50 Hz. This means it vibrates 50 times per second.

To calculate frequency when you know the total time and the number of oscillations, use this relationship: frequency = number of oscillations / time taken In symbols: f = N / t Here, f is frequency in hertz (Hz), N is the number of complete oscillations, and t is the time in seconds (s).

Because many objects vibrate very quickly, we often use metric prefixes to make the numbers easier to read. **Kilohertz** (kHz) means thousands of hertz. **Megahertz** (MHz) means millions of hertz. **Gigahertz** (GHz) means billions of hertz. - 1 kHz = 1,000 Hz (10^3 Hz) - 1 MHz = 1,000,000 Hz (10^6 Hz) - 1 GHz = 1,000,000,000 Hz (10^9 Hz)

Worked Example: A guitar string is plucked and vibrates. You count that it completes 240 full oscillations in 3 seconds. What is the frequency? Step 1: Identify the known values. Number of oscillations (N) = 240. Time (t) = 3 s. Step 2: Choose the correct formula. $f = N/t$. Step 3: Substitute the values into the equation. $f = 240/3$. Step 4: Calculate the result. $f = 80$. Step 5: State the answer with units. The frequency is 80 Hz.

Complete the sentence: Frequency is measured in hertz, where one hertz equals __________ oscillation per second.

Word bank: hertz · oscillations · kilohertz · megahertz · gigahertz

Activity

Measuring a Pendulum's Frequency

You will need: a clamp stand, a string, a small bob (like a metal washer or plasticine ball), and a stopwatch.

- Set up the pendulum by attaching the string to the clamp stand so it hangs freely. Measure the length of the string from the clamp to the centre of the bob.
- Pull the bob slightly to one side and let it go. This start point is your rest position reference.
- Start the stopwatch exactly as you release the bob. Count every time the bob returns to

the *same* starting side (this counts as one full oscillation).

- Stop the watch after counting exactly 10 complete oscillations. Record this time in the table below.
- Repeat the experiment two more times to get three results. Calculate the average time for 10 oscillations.

Trial	Time for 10 oscillations (s)
1	___________
2	___________
3	___________
Average	___________

Using your average time, calculate the frequency of the pendulum. Frequency = $10/\text{average time}$ Frequency = ___________ Hz

Questions

1. Define what is meant by one complete oscillation.

2. A tuning fork vibrates at 440 Hz. How many oscillations does it complete in 5 seconds?

3. An electron in a circuit completes 2,000,000 oscillations in 1 second. What is its frequency in megahertz (MHz)?

4. A pendulum takes 12 seconds to complete 6 oscillations. Calculate its frequency in hertz.

5. Explain why the length of the string might affect the frequency of the pendulum in the activity, even though we didn't measure it here.

Extension

Research the typical frequency range of human hearing (20 Hz to 20 kHz). Calculate how many oscillations occur in one minute for the lowest audible note (20 Hz) and the highest audible note (20,000 Hz). Compare these two numbers to understand the scale of vibration speeds involved.